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Vehicle behavior control device

Abstract

A vehicle behavior control device includes: an ith suspension mounted on an ith wheel of a vehicle (where i=1 to 3); a fourth suspension mounted on a fourth wheel; and a controller. The ith suspension includes an ith actuator. The controller converts a required value of a behavior parameter to an ith required control force for the ith actuator. When a kth required control force (k is any of 1 to 3) is greater than a kth output range according to an output capability of a kth actuator, the controller performs a required control force reduction process of reducing all of first to third required control forces, and controls the ith actuator based on the ith required control force after the required control force reduction process.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority to Japanese Patent Application No. 2023-131228 filed on Aug. 10, 2023, incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

(2) The present disclosure relates to techniques for controlling the behavior of a vehicle.

2. Description of Related Art

(3) Japanese Unexamined Patent Application Publication No. 2023-47810 (JP 2023-47810 A) discloses a technique for controlling the behavior of a vehicle using active suspensions. Each active suspension includes an actuator that can actively apply a control force in an up-down direction to a wheel. According to the technique described in JP 2023-47810 A, even when such actuators are mounted on only three wheels, it is possible to implement vehicle behavior control equivalent to the case where the actuators are mounted on four wheels.

SUMMARY

(4) An example in which vehicle behavior control is performed using three actuators as described in JP 2023-47810 A will be considered. When the output of an actuator reaches the upper limit of

the output of the actuator, it may cause a feeling of discomfort toward the vehicle behavior.
(5) A first aspect relates to a vehicle behavior control device.
(6) The vehicle behavior control device includes: an ith suspension mounted on an ith wheel of four wheels of a vehicle (where i=1 to 3); a fourth suspension mounted on a fourth wheel other than the ith wheel; and a controller.
(7) The ith suspension includes an ith actuator configured to apply a control force in an up-down direction to the ith wheel.
(8) The controller calculates a required value of a behavior parameter representing behavior of the vehicle.
(9) The controller converts the required value of the behavior parameter to an ith required control force for the ith actuator.
(10) The controller then controls the behavior of the vehicle by controlling the ith actuator based on the ith required control force.
(11) When a kth required control force (where k is any of 1 to 3) is greater than a kth output range according to an output capability of a kth actuator, the controller performs a required control force reduction process of reducing all of a first required control force, a second required control force, and a third required control force, and controls the ith actuator based on the ith required control force after the required control force reduction process.
(12) According to the present disclosure, the required control force reduction process is performed when the kth required control force for the kth actuator is greater than the kth output range according to the output capability of the kth actuator. All of the first to third required control forces for first to third actuators are reduced by the required control force reduction process. The ith actuator is then controlled based on the ith required control force after the required control force reduction process. This reduces the possibility of the output of the ith actuator reaching its output upper limit. As a result, a feeling of discomfort toward the vehicle behavior is less likely to be caused.
(13) As a result of the required control force reduction process, a period during which the kth required control force for the kth actuator is greater than the output upper limit of the kth actuator is reduced. Therefore, a period during which active vehicle behavior control cannot be performed is reduced. This results in improved comfort.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Features, advantages, and technical and industrial significance of exemplary embodiments of the disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:
(2) FIG. **1** is a schematic diagram illustrating a configuration example of a vehicle behavior control device;
(3) FIG. **2** is a conceptual diagram illustrating vehicle behavior control using three actuators;
(4) FIG. **3**A is a conceptual diagram illustrating an example of a required control force reduction process;
(5) FIG. **3**B is a conceptual diagram illustrating an example of the required control force reduction process;
(6) FIG. **3**C is a conceptual diagram illustrating an example of the required control force reduction process;
(7) FIG. **4** is a flowchart illustrating a first example of a process related to the vehicle behavior control;
(8) FIG. **5** is a flowchart illustrating a second example of the process related to the vehicle behavior

control; and
(9) FIG. **6** is a flowchart illustrating a third example of the process related to the vehicle behavior control.
DETAILED DESCRIPTION OF EMBODIMENTS
1. Configuration Example of Vehicle Behavior Control Device
(10) FIG. **1** is a schematic diagram illustrating a configuration example of a vehicle behavior control device that controls the behavior of a vehicle **10** according to the present embodiment. Vehicle **10** includes a left front wheel **14**FL and a right front wheel **14**FR in a front shaft **16**F. The vehicles **10** are provided with a left rear wheel **14**RL and a right rear wheel **14**RR in the rear shaft **16**R. The vehicle behavior control device includes a suspension **20**FL, **20**FRA, **20**RLA, **20**RRA, controllers **30**, and a sensor group **40**.
(11) The suspension **20**FL is provided with respect to the left front wheel **14**FL, and suspends the left front wheel **14**FL from the vehicle body **12**. The suspension **20**FL includes a spring **22**FL and a shock absorber **24**FL.
(12) The suspension **20**FRA is provided with respect to the right front wheel **14**FR, and suspends the right front wheel **14**FR from the vehicle body **12**. The suspension **20**FRA includes an actuator **26**FR in addition to the spring **22**FR and the shock absorber **24**FR. The actuator **26**FR is configured to actively apply a control force in an up-down direction to the right front wheel **14**FR. That is, the suspension **20**FRA is a fully active suspension.
(13) The suspension **20**RLA is provided for the left rear wheel **14**RL, and suspends the left rear wheel **14**RL from the vehicle body **12**. The suspension **20**RLA includes an actuator **26**RL in addition to the spring **22**RL and the shock absorber **24**RL. The actuator **26**RL is configured to actively apply a control force in the up-down direction to the left rear wheel **14**RL. That is, the suspension **20**RLA is a fully active suspension.
(14) The suspension **20**RRA is provided for the right rear wheel **14**RR, and suspends the right rear wheel **14**RR from the vehicle body **12**. The suspension **20**RRA includes an actuator **26**RR in addition to the spring **22**RR and the shock absorber **24**RR. The actuator **26**RR is configured to actively apply a control force in the up-down direction to the right rear wheel **14**RR. That is, the suspension **20**RRA is a fully active suspension.
(15) The configuration and the mechanism of the actuator **26** are not particularly limited. For example, the actuator **26** includes a motor supported by a sprung structure and a torsion bar coupled to an output shaft of the motor. An end portion of the torsion bar is connected to a suspension arm of the suspension **20** via a link mechanism. Rotation of the torsion bar by the motor is converted into movement in the up-down direction of the wheels **14** via a link mechanism. This allows the wheels **14** to be actively moved up and down.
(16) The suspension **20**FL provided for the left front wheel **14**FL is an inactive suspension without the actuator **26**.
(17) The controllers **30** are connected to the sensor group **40** via an in-vehicle network such as a Controller Area Network (CAN). The controller **30** acquires a signal from the sensor group **40**. The sensor group **40** includes, for example, a sensor that measures a physical quantity related to the behavior of the vehicle **10**, such as an acceleration sensor, a vehicle height sensor, and a wheel speed sensor. The controllers **30** are also connected to the actuator **26**RL, **26**RR, **26**FR via an in-vehicle network.
(18) The controller **30** includes a processor **32** and a memory **34** coupled to the processor **32**. The processor **32** performs various processes. The memory **34** stores a program **36** that is executable by the processor **32** and various types of information related to the program **36**. The program **36** may be recorded in a computer-readable recording medium. The functions of the controller **30** are implemented by cooperation of the processor **32** executing the program **36** and the memory **34**.
(19) The controller **30** performs “vehicle behavior control” for controlling the behavior of the vehicle **10**. Examples of the vehicle behavior control include sprung feedback control, unsprung

feedback control, attitude control, and preview control. The sprung feedback control suppresses vibration of the sprung member based on the sprung state quantity calculated by using the measurement value of the sprung acceleration sensor. The unsprung feedback control suppresses vibration of the unsprung member based on the unsprung state quantity calculated by using the measured values of the unsprung acceleration sensor and the vehicle height sensor. The attitude control controls the attitude with respect to steering and acceleration/deceleration. In the preview control, a road surface state is pre-read using a database of camera images and high-precision map data, and vibration is suppressed. These various controls may be combined.
(20) The controllers **30** perform desired vehicle behavior control by controlling the actuator **26**FR, **26**RL, **26**RR of the suspension **20**FR, **20**RL, **20**RR based on the signals obtained by the sensor group **40**. Although the actuator **26** is not provided for the left front wheel **14**FL, the same controllability as when the actuator **26** is provided for all of the four wheels can be achieved by only the three actuator **26**FR, **26**RL, **26**RR. The vehicle behavior control by the three actuator **26**FR, **26**RL, **26**RR will be described in more detail below.
2. Vehicle Behavior Control
(21) The vehicle behavior control by the three actuator **26**FR, **26**RL, **26**RR is the same as that described in, for example, JP 2023-47810 A.
(22) FIG. **2** shows a behavior model for vehicle behavior control. The behavior of the vehicle **10** is represented by behavior parameters. In the behavioral model shown in FIG. **2**, the behavioral parameters include the mode of motion at the sprung center of gravity position of the vehicle **10**, i.e., the roll moment M.sub.r, the pitch moment M.sub.p, and the heave force F.sub.h. Hereinafter, motion modes consisting of a roll moment M.sub.r, a pitch moment M.sub.p, and a heave force F.sub.h will be referred to as three center-of-gravity modes. The controller **30** first converts the required control force required for any type of vehicle behavior control into the required values of the three center-of-gravity modes (behavior parameters).
(23) For example, for a certain vehicle behavior control, a required control force in the up-down direction is required for each of the four wheels of the vehicle **10**. Specifically, a required control force F.sub.fli in the up-down direction with respect to the left front wheel **14**FL, a required control force F.sub.fri in the up-down direction with respect to the right front wheel **14**FR, a required control force F.sub.rli in the up-down direction with respect to the left rear wheel **14**RL, and a required control force F.sub.rri in the up-down direction with respect to the right rear wheel **14**RR are required. The controller **30** converts the required control forces F.sub.fli, F.sub.fri, F.sub.rli, and F.sub.rri for these four wheels into the required values of the three center-of-gravity modes according to Expression (1) below.

(24) (Expression1)

$$\begin{bmatrix} F_h \\ M_r \\ M_p \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ \frac{-T_f}{2} & \frac{T_f}{2} & \frac{-T_r}{2} & \frac{T_r}{2} \\ -l_f & -l_f & l_r & l_r \end{bmatrix} \begin{bmatrix} F_{fri} \\ F_{fli} \\ F_{rri} \\ F_{rli} \end{bmatrix} \quad (1)$$

(25) In Expression (1), lf, lr, Tf, and Tr are a distance between the front shaft **16**F and the center of gravity, a distance between the rear shaft **16**R and the center of gravity, a front tread, and a rear tread, respectively (see FIG. **2**). These parameters are given in advance as specification information of the vehicle **10**. The controller **30** computes the required values (F.sub.h, M.sub.r, M.sub.p) of the three center-of-gravity modes based on the specifications information of the vehicle **10** and the required control forces (F.sub.fli, F.sub.fri, F.sub.rli, F.sub.rri) for the four wheels. The required values of the three center-of-gravity modes may include a roll moment, a pitch moment, and a heave force required for attitude control accompanying steering and acceleration/deceleration.
(26) After calculating the required values F.sub.h, M.sub.r, and M.sub.p of the three center-of-gravity modes, the controller **30** converts the required values F.sub.h, M.sub.r, and M.sub.p of the three center-of-gravity modes to required control forces F.sub.fr, F.sub.rr, and F.sub.rl for the three wheels **14**FR, **14**RR, and **14**RL according to Expression (2) below.

$$\begin{bmatrix} F_{fr} \\ \cdots \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ -T_f & -T_r & T_r \\ \cdots \end{bmatrix}^{-1} \begin{bmatrix} F_h \\ M \cdots \end{bmatrix} \quad (2)$$

(27) (Expression2) $F_{rl} \quad -l_f \quad l_r \quad l_r \quad M_p$

(28) The required control forces F.sub.fr, F.sub.rr, and F.sub.rl for the three wheel **14**FR, **14**RR, and **14**RL correspond to the required control forces for the three actuator **26**FR, **26**RR, and **26**RL. The controller **30** controls the actuator **26**FR so that the control force in the up-down direction applied to the right front wheel **14**FR becomes the required control force F.sub.fr. Similarly, the controller **30** controls the actuator **26**RR so that the control force in the up-down direction applied to the right rear wheel **14**RR becomes the required control force F.sub.rr. Similarly, the controller **30** controls the actuator **26**RL so that the control force in the up-down direction applied to the left rear wheel **14**RL becomes the required control force F.sub.rl.
(29) By distributing the required values of the three center-of-gravity modes to the three actuators **26**FR, **26**RR, and **26**RL in this manner, desired behavior including all of the roll, pitch, and heave is implemented in the vehicle **10**. That is, the desired vehicle behavior can be implemented by only the three actuators **26**FR, **26**RR, and **26**RL. It is preferable that the required actuator can be reduced from the viewpoints of improvement in mountability, reduction in cost, reduction in weight, power saving, and the like.
(30) The control method of the actuator **26** may be position control (angle control) instead of force control (torque control). Then, the controller **30** calculates the required position control amounts for the three actuators **26**FR, **26**RR, and **26**RL such that the required control forces F.sub.fr, F.sub.rr, and F.sub.rl for the three actuators **26**FR, **26**RR, and **26**RL represented by the above Expression (2) are obtained. The controllers **30** perform position control of the actuator **26**FR, **26**RR, **26**RL in accordance with the required position control amounts.
(31) In the above explanation, the suspension **20**FL provided for the left front wheel **14**FL is an inactive suspension without the actuator **26**. However, the present embodiment is not limited thereto. Instead of the left front wheel **14**FL, the suspension **20** for the other wheels **14** may be an inactive suspension.
(32) The generalization is as follows: I takes a value from 1 to 3. The ith wheel **14-***i* is provided with an ith suspension **20-***i* that is an active suspension. The ith suspension **20-***i* includes an ith actuator **26-***i* that applies a control force in the up-down direction to the ith wheel **14-***i*. The fourth wheel **14-4** is provided with a fourth suspension **20-4** that is an inactive suspension. The fourth suspension **20-4** does not include the actuator **26**.
(33) The controller **30** calculates required values of the behavior parameters representing the behavior of the vehicle **10** (see Expression (1)). Subsequently, the controllers **30** convert the required values of the behavior parameters into the ith required control force Fi for the ith actuator **26-***i* (see Expression (2)). Then, the controllers **30** control the behavior of the vehicles **10** by controlling the ith actuator **26-***i* based on the ith required control force Fi. It is preferable in terms of improvement in mountability, reduction in cost, reduction in weight, power saving, etc. that the desired vehicle behavior can be implemented by only three actuators **26-***i* (i=1 to 3).
3. Required Control Force Reduction Process
(34) Consider a case where the output of an actuator **26** reaches the output upper limit of the actuator **26**. For convenience, the actuator **26** is referred to as a jth actuator **26-***j*. j is any of 1 to 3.
(35) In the four-actuator configuration, even if the output of the jth actuator **26-***j* reaches the output upper limit, the control effectiveness for the jth wheel **14-***j* is only reduced. However, in the three-actuator configuration as in the present embodiment, the jth actuator **26-***j* also covers the control of the fourth wheel **14-4** in which the actuator **26** is not provided. Therefore, when the output of the jth actuator **26-***j* reaches the output upper limit, there is a possibility that a sense of discomfort to the vehicular behavior may occur. For example, consider that there is no actuator **26** in the left front wheel **14**FL, and the upper limit of the output of the actuator **26**FR with respect to the right front wheel **14**FR is smaller than the actuator **26**RL, **26**RR with respect to the rear wheel **14**RL, **14**RR. When a large road surface input is applied to the left front wheel **14**FL, the output of the actuator

26FR to the right front wheel **14**FR is insufficient. As a result, the right front wheel **14**FR will swing although it is a road surface input to the left front wheel FL. This causes a sense of discomfort.
(36) In addition, active-vehicle behavior control cannot be performed while the jth required control force **26-***j* for the jth actuator Fj is greater than the output upper limit of the jth actuator **26-***j*, and thus comfort is reduced. In particular, in the case of a torsion bar active suspension, the stiffness of the torsion bar increases the wheel rate. Therefore, if the vehicle behavior control is not performed, the body vibration tends to increase. If such a state continues, the comfort is greatly reduced.
(37) In order to solve the above-described problem, the controller **30** according to the present embodiment performs a “required control force reduction process” as necessary.
(38) The operation condition for the required control force reduction process is that the kth required control force Fk for the kth actuator **26-***k* is greater than the kth output range according to the output capability of the kth actuator **26-***k*. k is at least one of 1 to 3. The kth output range according to the output capability of the kth actuator **26-***k* does not necessarily have to correspond to the maximal output possible range of the kth actuator **26-***k*. For example, the kth output range may be 70% or the like of the maximum possible output range. Information regarding the kth output range is given to the controller **30** in advance. The controller **30** determines whether the operation condition for the required control force reduction process is satisfied. When the operation condition is satisfied, the output of the kth actuator **26-***k* may reach the output upper limit, and therefore the controller **30** performs the required control force reduction process. Specifically, the controllers **30** reduce all of the first to third required control forces F**1** to F**3** for the first to third actuators **16-1** to **26-3**. Then, the controller **30** controls the behavior of the vehicle **10** by controlling the ith actuator **26-***i* based on the ith required control force Fi′ after the required control force reduction process.
(39) FIGS. **3**A to **3**C are conceptual diagrams illustrating examples of the required control force reduction process. The limits of the ith output range of the ith actuator **26-***i* (i=1-3) are represented by ith thresholds Thi. The ith thresholds Thi do not necessarily have to correspond to the maximal power of the ith actuator **26-***i*. For example, the ith thresholds Thi may be 70% of the maximal power, etc. The output ranges (i.e., first threshold Th**1** to third threshold Th**1**) for the first to third actuators **26-1** to **26-3** depend on the respective actuator capabilities and are independent of each other.
(40) In the illustrated **3**A, the first required control force F**1** for the first actuator **26-1** is greater than the first threshold Th**1** (first output range). Since the operation condition is satisfied, the required control force reduction process is performed. The controller **30** calculates first to third required control forces F**1**′ to F**3**′ by multiplying each of the first to third required control forces F**1** to F**3** by a correction factor α (0<α<1). The correction factor α may be set to be smaller as the difference between the first required control force F**1** and the first threshold Th**1** increases.
(41) Also in FIG. **3**B, the first required control force F**1** for the first actuator **26-1** is greater than the first threshold Th**1** (first output range). In the present embodiment, the required control force reduction process is performed such that all of the first to third required control forces F**1**′ to F**3**′ after the required control force reduction process fall within the first to third output ranges, respectively. The correction factor α can be set based on a ratio (F**1**/Th**1**) of the first required control force F**1** to the first threshold Th**1**.
(42) In FIG. **3**C, the first to third required control forces F**1** to F**3** for the three actuators **26-1** to **26-3** are greater than the first to third thresholds Th**1** to Th**3**, respectively. Also in this example, the required control force reduction process is performed such that all of the first to third required control forces F**1**′ to F**3**′ after the required control force reduction process fall within the first to third output ranges, respectively. The correction factor α can be set based on the highest one of the three ratios F**1**/Th**1**, F**2**/Th**2**, and F**3**/Th**3**.
(43) Alternatively, the required control force reduction process may apply a high-pass filter to the first to third required control forces F**1** to F**3**. Since the required control force tends to be greater

for the lower frequency component, the required control force can be reduced by applying the high-pass filter. In FIG. **3**A, the first required control force F**1** for the first actuator **26-1** is greater than the first threshold Th**1** (first output range). In this case, the larger the difference between the first required control force F**1** and the first threshold Th**1** is, the higher the filtering effect may be. By increasing the cutoff frequency of the high-pass filter or increasing the order of the high-pass filter, the filter effect can be enhanced.
(44) As yet another example, the required control force reduction process may reduce the control gain of the vehicle behavior control. More specifically, the vehicle behavior control includes first vehicle behavior control for a road surface input and second vehicle behavior control for an operation input (steering, acceleration, or deceleration). The controller **30** calculates required values of the behavior parameters based on the first control gain G**1** of the first vehicle behavior control and the second control gain G**2** of the second vehicle behavior control. In the required control force reduction process, the controller **30** performs a "gain reduction process" of reducing either or both of the first control gain G**1** and the second control gain G**2**. Then, the controller **30** calculates the required values of the behavior parameters again based on the first control gain G**1**′ and the second control gain G**2**′ after the gain reduction process. This can also reduce all of the first to third required control forces F**1** to F**3** for the first to third actuators **26-1** to **26-3**.
(45) Various examples of the gain reduction processing are conceivable. For example, the controller **30** may multiply each of the first control gain G**1** and the second control gain G**2** by a correction factor β (0<β<1).
(46) As another example of the gain reduction process, the priorities of the first vehicle behavior control and second vehicle behavior control may be considered. More specifically, the first correction factor β1 is greater than 0 and equal to or less than 1 (0<β1≤1), and decreases as the priority of the first vehicle behavior control decreases. The second correction factor β2 is greater than 0 and equal to or less than 1 (0<β2≤1), and decreases as the priority of the second vehicle behavior control decreases. The controller **30** multiplies the first correction factor β1 by the first control gain G**1**, and multiplies the second correction factor β2 by the second control gain G**2**. The correction factor for the first vehicle behavior control or the second vehicle behavior control, whichever has higher priority, may be set to 1.0. Then, only the first control gain G**1** or the second control gain G**2** decreases.
(47) The first control gain G**1** of the first vehicle behavior control may be further divided into a feedback gain and a feedforward gain, and the priorities of the feedback gain and the feedforward gain may be considered. The second control gain G**2** of the second vehicle behavior control may be further divided into a steering input gain and an acceleration/deceleration input gain, and the priorities of the steering input gain and the acceleration/deceleration input gain may be considered.
(48) Two or more of the above examples of the required control force reduction process may be combined.
(49) As described above, according to the present embodiment, when the kth required control force Fk for the kth actuator **26-***k* is greater than the kth output range according to the output capability of the kth actuator **26-***k*, the required control force reduction process is performed. The required control force reduction process reduces all of the first to third required control forces F**1** to F**3** for the first to third actuators **26-1** to **26-3**. The ith actuator **26-***i* is then controlled based on the ith required control force Fi′ after the required control force reduction process. This reduces the likelihood that the output of the ith actuator **26-***i* reaches the output upper limit. As a result, the occurrence of a sense of discomfort with respect to the vehicle behavior is suppressed.
(50) As a result of the required control force reduction process, the period during which the kth required control force Fk for the kth actuator **26-***k* is greater than the output upper limit of the kth actuator **26-***k* is reduced. Therefore, the period during which the active vehicle behavior control cannot be performed is shortened. As a result, comfort is improved.
(51) Preferably, the required control force reduction process is performed such that the ith required

control force Fi′ after the required control force reduction process falls within the ith output range according to the output capability of the ith actuator. In this case, the above-mentioned effects are further increased.

4. Example of Process Flow

(52) Hereinafter, some examples of the process flow related to the vehicle behavior control according to the present embodiment will be described.

4-1. First Example

(53) FIG. **4** is a flowchart illustrating a first example of a process related to vehicle behavior control. In S**100**, the controllers **30** calculate required values of the behavior parameters representing the behavior of the vehicles **10** (see Expression (1)). In S**200**, the controller **30** converts the required values of the behavior parameters to the first to third required control forces F**1** to F**3** for the first to third actuators **26-1** to **26-3** (see Expression (2)).

(54) In the subsequent step S**300**, the controller **30** determines whether the operation condition for the required control force reduction process is satisfied. The operation condition for the required control force reduction process is that the kth required control force Fk for the kth actuator **26-***k* is greater than the kth output range (kth threshold Thk) according to the output capability of the kth actuator **26-***k*.

(55) When the operating condition is not satisfied (S**300**; No), the process proceeds to S**500**. In S**500**, the controllers **30** perform the vehicle behavior control by controlling the ith actuator **26-***i* based on the ith required control force Fi.

(56) On the other hand, when the operating condition is satisfied (S**300**; Yes), the process proceeds to S**400**. In S**400**, the controllers **30** perform the required control force reduction process. The required control force reduction process is as described in Section 3 above. Thereafter, the process proceeds to S**500**. In S**500**, the controllers **30** perform the vehicle behavior control by controlling the ith actuator **26-***i* based on the ith required control force Fi′ after the required control force reduction process.

4-2. Second Example

(57) FIG. **5** is a flowchart illustrating a second example of a process related to vehicle behavior control. The second example is the same as the first example described above except for S**400**. In S**400**, the controllers **30** perform the required control force reduction process. In this required control force reduction process, it is not checked whether all of the first to third required control force F**1**′ to F**3**′ after the required control force reduction process fall within the first to third output ranges, respectively. Instead, after S**400**, the process returns to S**300**. By repeating S**300** and S**400**, it is ensured that all of the first to third required control forces F**1**′ to F**3**′ after the required control force reduction process fall within the first to third output ranges, respectively.

4-3. Third Example

(58) FIG. **6** is a flowchart illustrating a third example of a process related to vehicle behavior control. The vehicle behavior control includes first vehicle behavior control for a road surface input and second vehicle behavior control for an operation input (steering, acceleration, or deceleration).

(59) In S**100**, the controller **30** calculates the required values of the behavior parameters based on the first control gain G**1** of the first vehicle behavior control and the second control gain G**2** of the second vehicle behavior control. S**200** and S**300** are the same as in the first example. When the operation condition for the required control force reduction process is satisfied (S**300**; Yes), the process proceeds to S**410**.

(60) In S**410**, the controller **30** performs a gain reduction process of reducing either or both of the first control gain G**1** and the second control gain G**2**. Various examples of the gain reduction process are as described above. After S**410**, the process returns to S**100**. In S**100**, the controller **30** calculates the required values of the behavior parameters again based on the first control gain G**1**′ and the second control gain G**2**′ after the gain reduction process. In a subsequent S**200**, the controller **30** converts the required values of the behavior parameters from the first to third required

control forces F**1** to F**3** for the first to third actuators **26-1** to **26-3**.
(61) In this way, it is also possible to reduce all of the first to third required control forces F**1** to F**3** by the gain reduction process. It can also be said that the combination of S**410**, S**100**, and S**200** is the required control force reduction process (S**400**).

Claims

1. A vehicle behavior control device, comprising: an ith suspension mounted on an ith wheel of four wheels of a vehicle (where i=1 to 3); a fourth suspension mounted on a fourth wheel other than the ith wheel; and a controller, wherein the ith suspension includes an ith actuator configured to apply a control force in an up-down direction to the ith wheel, the controller is configured to calculate a required value of a behavior parameter representing behavior of the vehicle, convert the required value of the behavior parameter to an ith required control force for the ith actuator, and control the behavior of the vehicle by controlling the ith actuator based on the ith required control force, and the controller is configured to, when a kth required control force (where k is any of 1 to 3) is greater than a kth output range according to an output capability of a kth actuator, perform a required control force reduction process of reducing all of a first required control force, a second required control force, and a third required control force, and control the ith actuator based on the ith required control force after the required control force reduction process.
2. The vehicle behavior control device according to claim 1, wherein the required control force reduction process includes either or both of multiplying each of the first required control force, the second required control force, and the third required control force by a correction factor of less than one, and applying a high-pass filter to the first required control force, the second required control force, and the third required control force.
3. The vehicle behavior control device according to claim 1, wherein the controller is configured to perform the required control force reduction process in such a manner that the ith required control force after the required control force reduction process is within an ith output range according to an output capability of the ith actuator.
4. The vehicle behavior control device according to claim 1, wherein control of the behavior of the vehicle includes first vehicle behavior control for a road surface input and second vehicle behavior control for steering, acceleration, or deceleration, the controller is configured to calculate the required value of the behavior parameter based on a first control gain of the first vehicle behavior control and a second control gain of the second vehicle behavior control, and the required control force reduction process includes a gain reduction process of reducing either or both of the first control gain and the second control gain, and calculating the required value of the behavior parameter again based on the first control gain and the second control gain after the gain reduction process.
5. The vehicle behavior control device according to claim 4, wherein the gain reduction process is either multiplying each of the first control gain and the second control gain by a correction factor of less than one, or multiplying the first control gain by a first correction factor and multiplying the second control gain by a second correction factor, the first correction factor being a correction factor that decreases as priority of the first vehicle behavior control decreases, and the second correction factor being a correction factor that decreases as priority of the second vehicle behavior control decreases.